

Physical Pharmacy

Code: (CPT 203, PT 203)

Lecture 6

SURFACE ACTIVE AGENT (SURFACTANT)

Lecture Learning Outcomes (LOs)

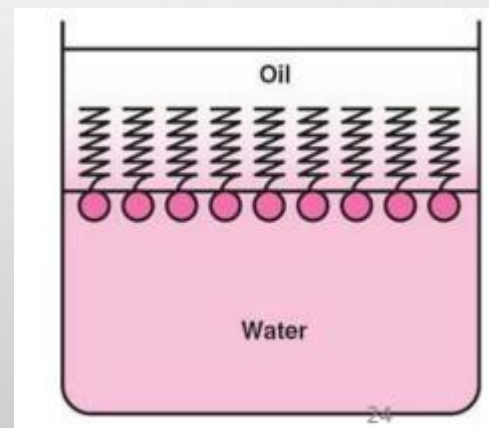
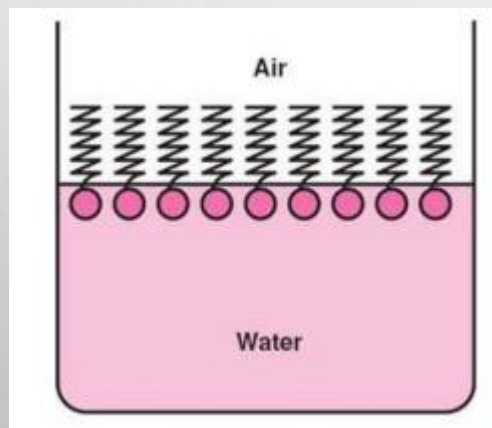
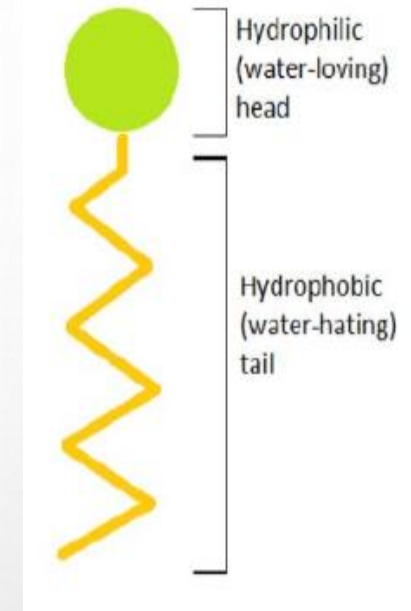
Program Outcomes (NARS 2017)		Course Learning Outcomes	
Code	Text	Code	Text
1-1-1	Demonstrate understanding of knowledge of pharmaceutical, biomedical, social, behavioral, administrative, and clinical sciences.	a1	Recognize different physicochemical properties affecting the behavior of different types of materials (solid, liquid and gas).
1.1.3	Integrate knowledge from fundamental sciences to handle, identify, extract, design, prepare, analyze, and assure quality of synthetic/natural pharmaceutical materials/products.	a2	Classify different factors concerning physicochemical properties such as solubility , viscosity, pH, isotonicity, dissolution, partition coefficient, surface tension and others for optimizing pharmaceutical formulations
4.2.2	Use contemporary technologies and media to demonstrate effective presentation skills.	d1	Provide efficient presentation using modern technology and media

Surface active agent (SAA) (Surfactant)

Surfactants have two distinct regions in their chemical structure, one of which is water-liking (hydrophilic) and the other of which is lipid-liking (lipophilic). These molecules are referred to as amphiphile.

THE BALANCED AMPHIPHILIC NATURE OF S.A.A. IS THE CHARACTER THAT CAUSE THEM TO BE ADSORBED AT THE INTERFACES (LIQUID/GAS OR LIQUID/LIQUID).

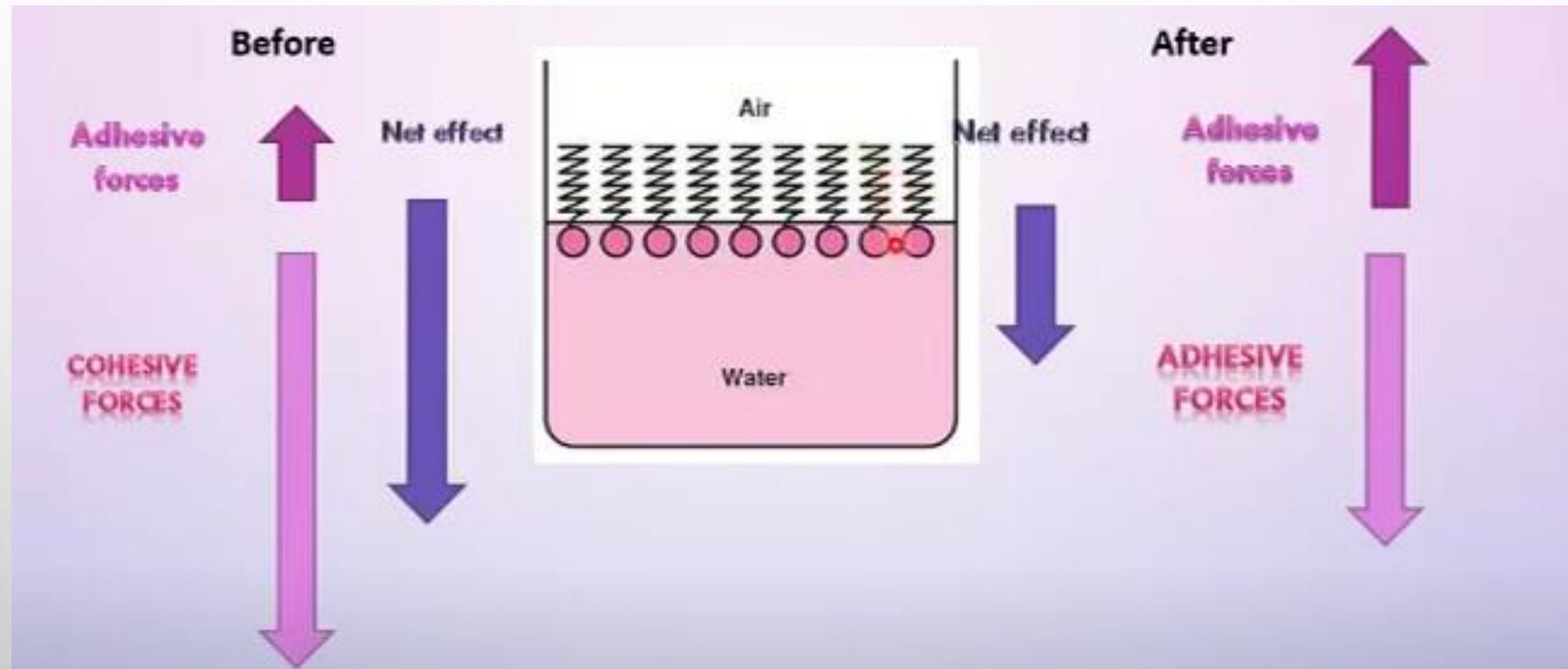
WHEN SUCH MOLECULE IS PLACED IN AN AIR-WATER OR OIL-WATER SYSTEM, THE POLAR GROUPS ARE ORIENTED TOWARD THE WATER, AND THE NONPOLAR GROUPS ARE ORIENTED TOWARD THE AIR OR OIL.



What is the role of a Surfactant



After addition of surfactant



Difference between the two applied forces on surface molecules

surface tension

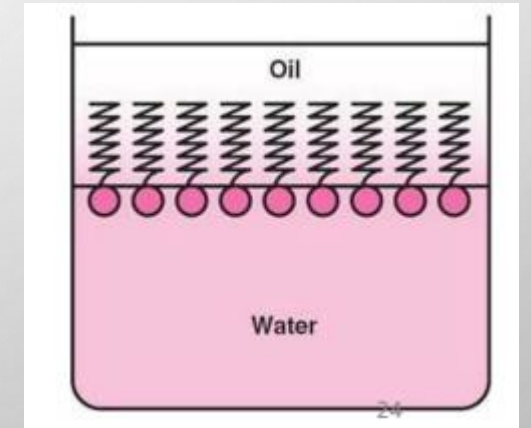
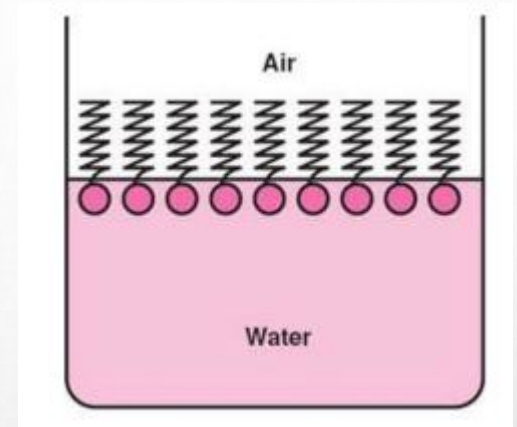
Upon addition of SAA it will be adsorbed at the surface (or interface) leading to a reduction in surface tension (or interfacial tension) of the liquid upon which it adsorbed



During this adsorption the SAA molecules present as discrete monomer



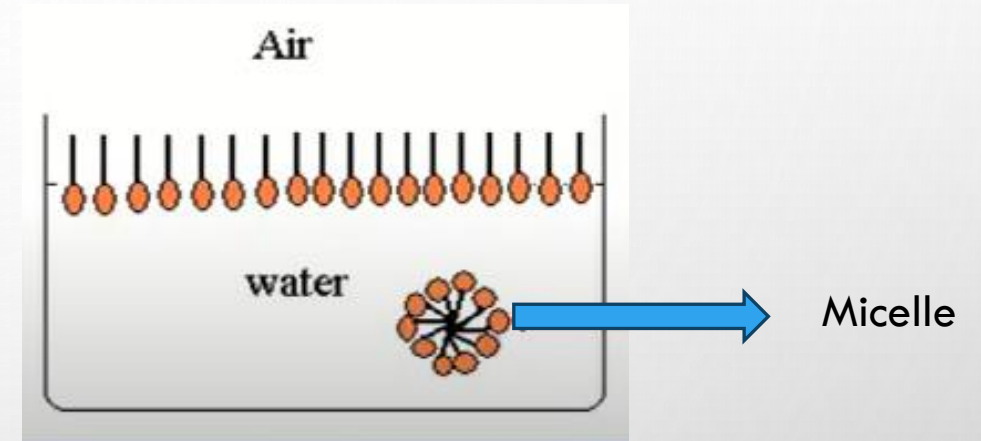
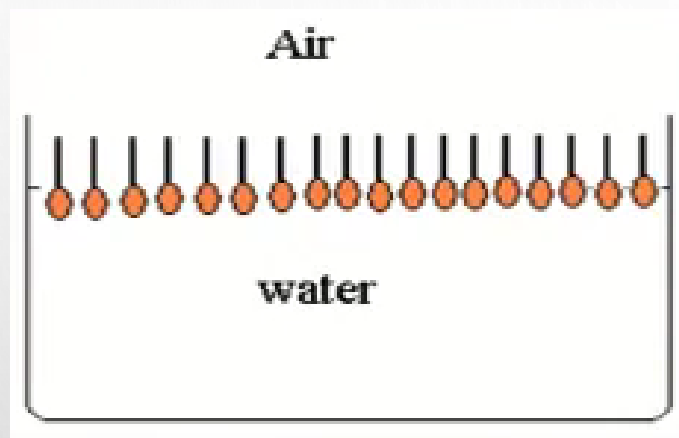
The SAA adsorption will continue with further addition of SAA till saturation of liquid surface occurs



What will happen after the surface saturation with further addition of SAA



After Saturation of liquid surface with continuing addition of SAA, micelles will be formed in the bulk of the liquid



Micelles: they are aggregates of SAA molecules formed in the bulk of the liquid after the surface saturation

- ✓ Micelles are dynamic structures and are continually formed and broken down in solution (they are not solid spheres).
- ✓ The main reason for micelle formation is to obtain a minimum free energy state.
- ✓ In a micelle, polar or ionic heads form an outer shell in contact with water, while non polar tails are sequestered in the interior to avoid water.



The point at which micelles begins to form is called Critical Micellar Concentration (CMC)

Micelles shapes

Normal spherical micelles

The polar groups of the surfactant are in the periphery and the hydrocarbon chains are oriented toward the center forming the core of the micelle

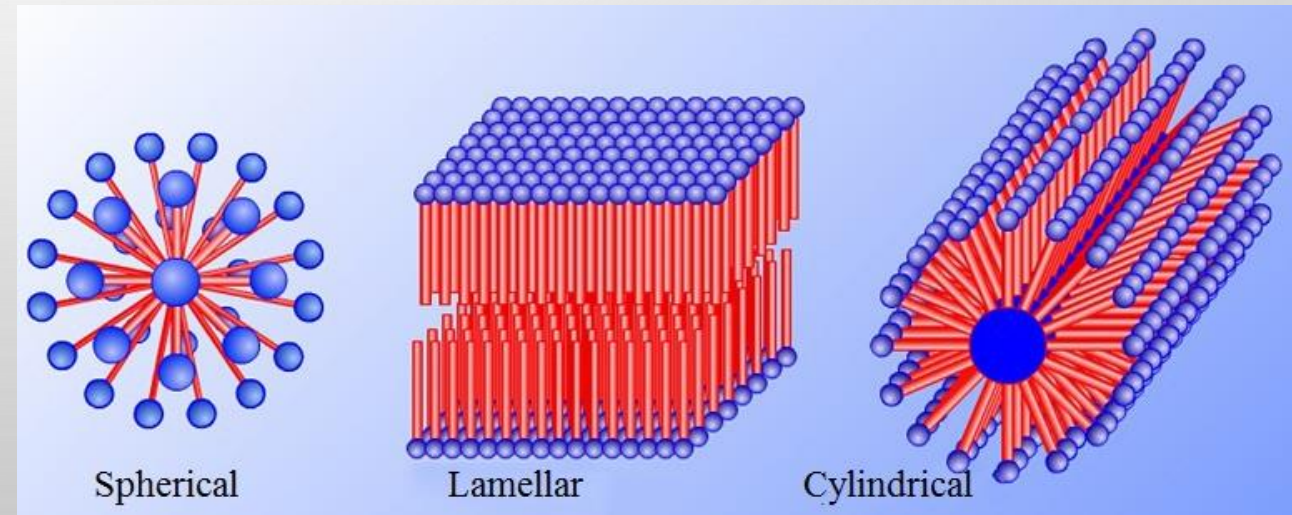
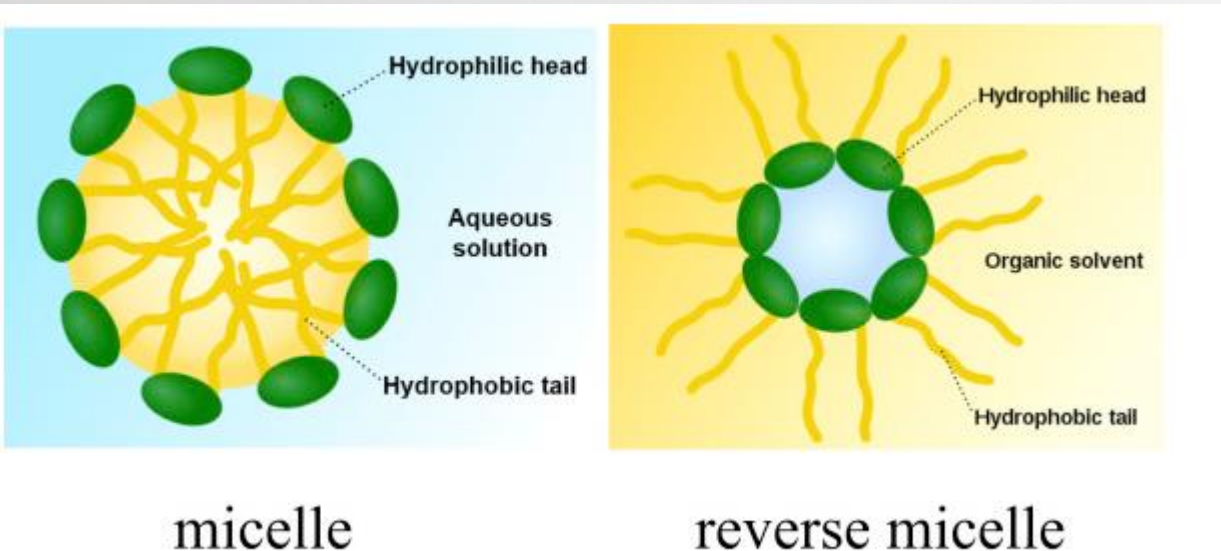
Inverted spherical micelles

In solvents of low polarity or oils ---- micelles are inverted

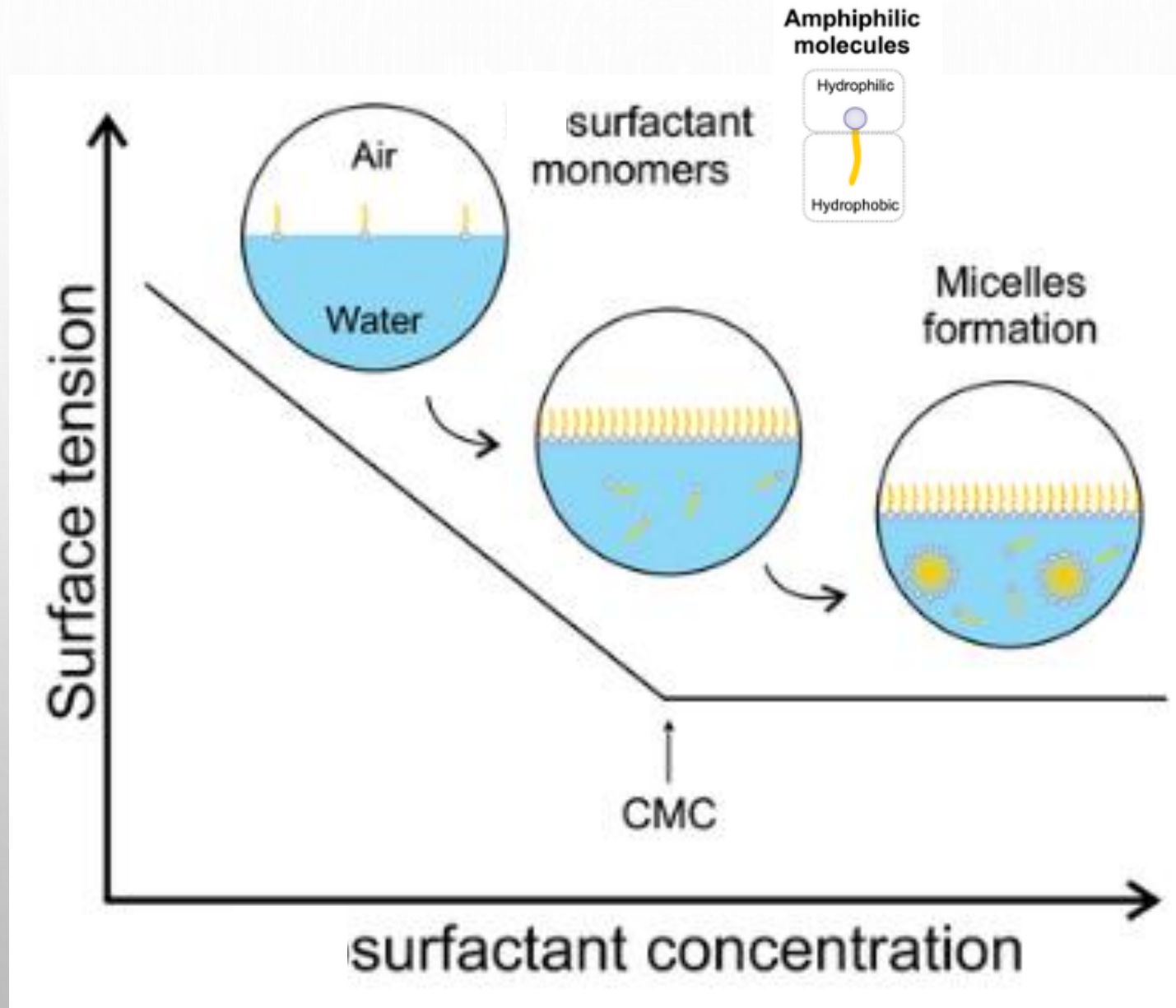
The polar groups face inward to form the core of the micelle while the hydrocarbon chains are oriented outward

Cylindrical and lamellar micelles

In more concentrated solutions of surfactant, micelles form these shapes



What happens to the surface tension after micelle formation



Surface tension before CMC:

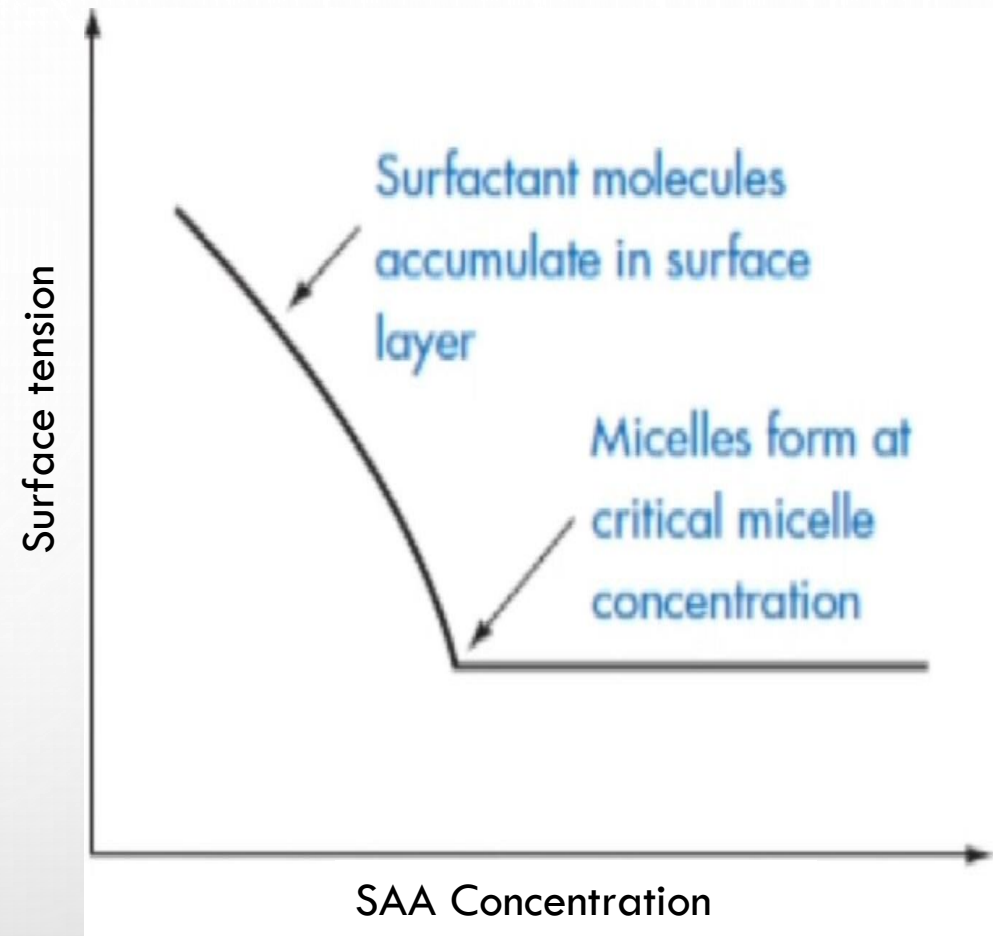
- ✓ Appreciable lowering of surface tension even at low concentrations.
- ✓ As the surfactant concentration **is increased**, the surface tension **continues to decrease** till critical micelle concentration (CMC).

CMC:

- ✓ Formation of small spherical aggregates or micelles in the bulk is considered as **a mean of shielding the hydrophobic portions from the aqueous environment**
- ✓ CMC corresponds to the concentration of SAA at which there is an abrupt change of slope of the plot.
- ✓ The slope becomes approximately zero at higher SAA concentrations.

Surface tension After CMC:

- ✓ An equilibrium is established between the surfactant molecules at the surface or interface and those remaining in the bulk of the solution.
- ✓ A partitioning is present between monomers in micelles and those in bulk and surface.
- ✓ When the concentration of the surfactant is increased above the CMC, the number of micelles increases but surface tension stays constant at the CMC value



Factors Affecting Micellization

1. Structure of the surfactant

- ✓ **Increase in length of the hydrocarbon chain results in a decrease in CMC**
(Facilitate the transfer from aqueous phase to micellar form)
- ✓ **Branched of hydrocarbon chain and the presence of double (unsaturated) bond causes an increase in CMC**
(Since the chains must come together inside the micelle)
- ✓ **Ionized surfactant have higher CMC in polar solvents compared to nonionic surfactant**
Greater interaction with water will retard micelle formation
The charges in the micelle of ionic surfactant are in close contact (Repulsion)

2. Addition of electrolytes

Electrolyte addition to solutions of ionic surfactants **decreases the CMC**

(This is because the electrolyte reduces the forces of repulsion between the charged head groups at the micelle surface)

In polar solvents:

↑ Polarity of Surfactant molecules → ↑ CMC

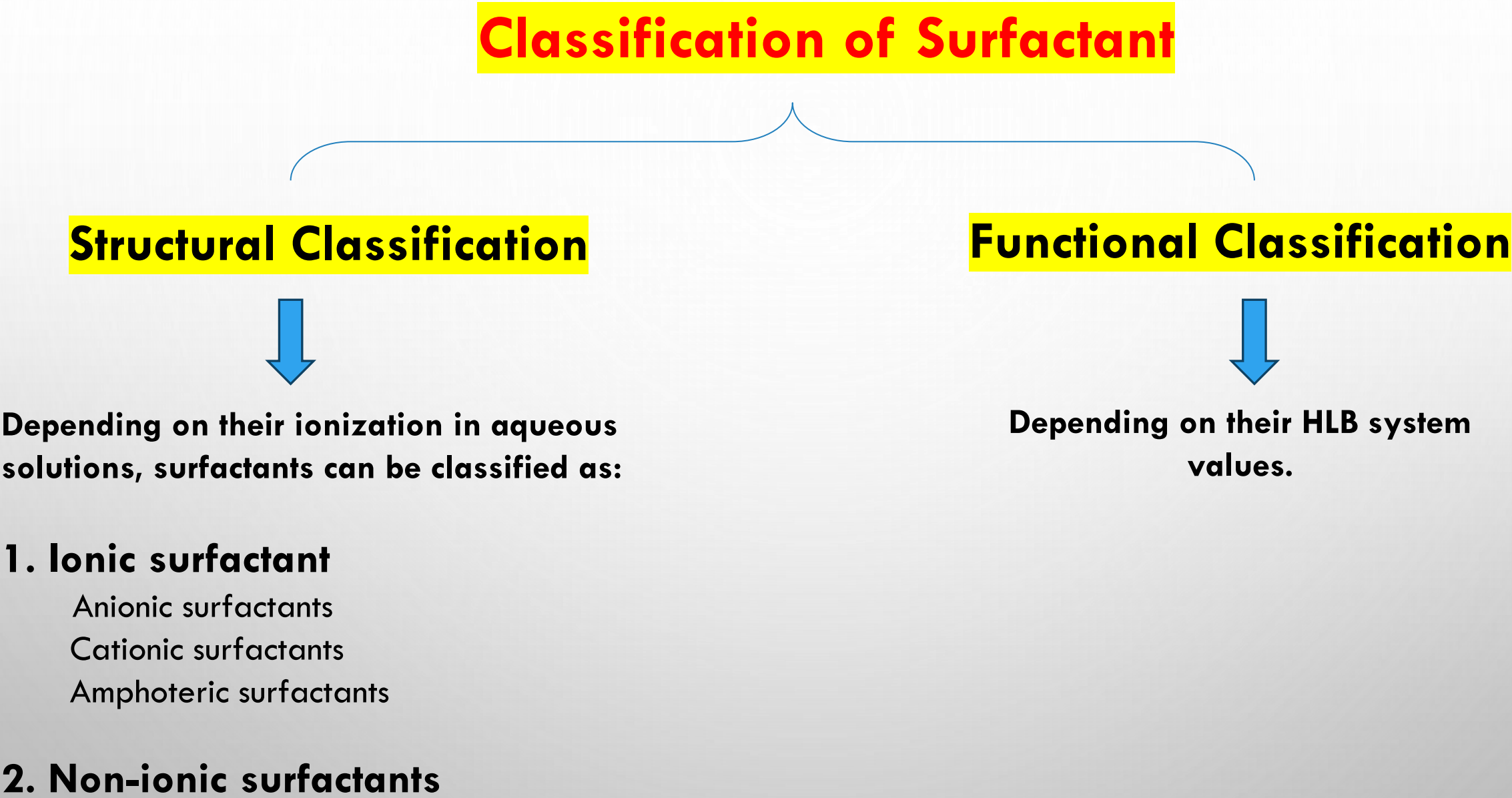
↑ Length of hydrocarbon chain → ↓ CMC

In non-polar solvents:

↑ Polarity of Surfactant molecules → ↓ CMC

↑ Length of hydrocarbon chain → ↑ CMC

Classification of Surfactant



Structural Classification



Depending on their ionization in aqueous solutions, surfactants can be classified as:

1. Ionic surfactant

- Anionic surfactants
- Cationic surfactants
- Amphoteric surfactants

2. Non-ionic surfactants

Functional Classification



Depending on their HLB system values.

Structural Classification



Non-ionic



Cationic



Anionic



Amphoteric



Non-ionic group



Hydrophobic group



Cationic head



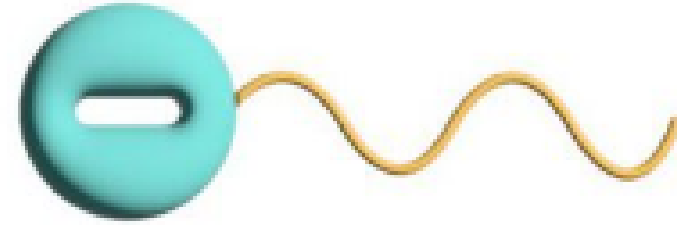
Anionic head



Zwitterionic head

Anionic Surfactants

Anionic S.A.A.: are effective at pH value > 8

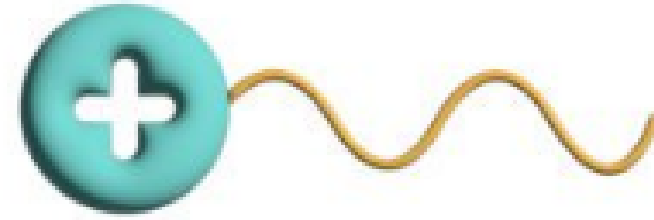


- ✓ Anionic Surfactant contain anionic functional group at their head, such as sulfonate, phosphate and carboxylates.
- ✓ More than 50% of the global usage of surfactant
- ✓ Use: 50% overall industrial production, laundry detergent, dish washing liquids.
- ✓ These agents are not suitable for internal use because of their unpleasant taste and irritant action on the intestinal mucosa (Toxic).

1. Alkali Soaps (Potassium stearate)
2. Sulphated compound such as sodium lauryl sulphate
3. Sulphonated compound such as sodium cetylsulphonate

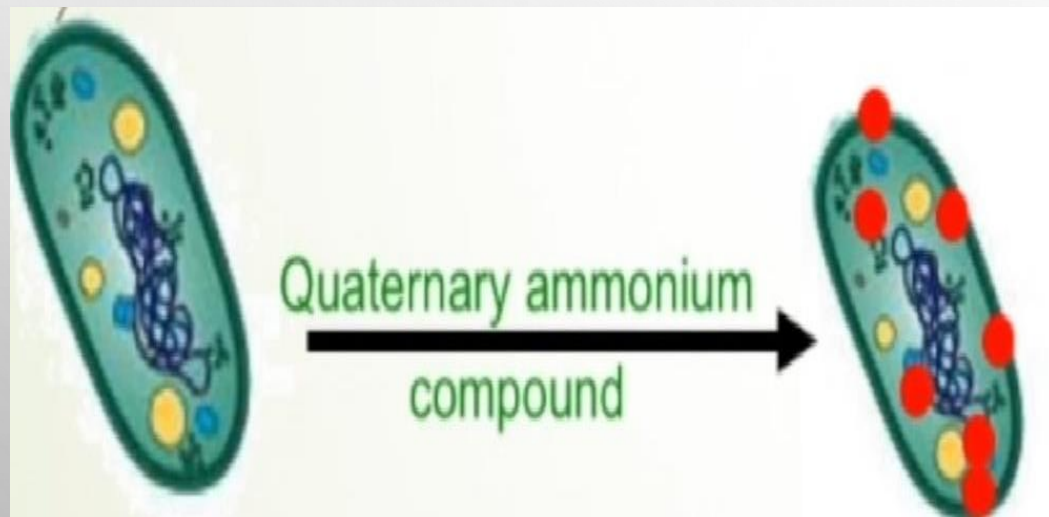
Cationic Surfactants

Cationic S.A.A.: are effective at pH of 3-7



- ✓ Like the anionic surfactants, the cationic ones required to be ionized to be effective.
- ✓ Are mainly quaternary ammonium compounds.
- ✓ They are more popular as antiseptics or disinfectants agents.
- ✓ They may be used on the skin, especially in the cleaning of wounds.
- ✓ They have bactericidal activity against a wide range of gram positive and some gram-negative organisms.

They have marked bactericidal activity against M.O. they can combine with (COOH) in the cell membrane of M.O. by cation exchange causing **increase in cell permeability and cell lysis.**



Adsorbed of SAA
on the cell surface

Increase the permeability or
leakiness of the lipid cell
membrane leading to cell lysis
(death)

Amphoteric Surfactants



Ampholytic surfactants are substances whose ionic characteristics depend on the pH of the system.

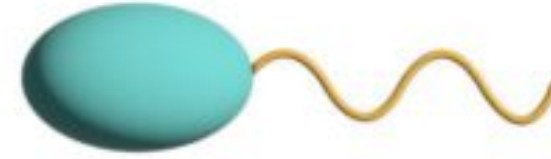
- ✓ when in **acidic solutions** these surfactants will be **positively charged**.
- ✓ when in **alkaline solutions** these surfactants will be **negatively charged**
- ✓ When in **a neutral solution**, they will have a neutral charge behave as **zwitterions**..

These amphoteric surfactants are often used in personal care products like shampoos, cosmetics, and toiletries.

The variable charge on the molecules allows for a multitude of applications and uses.

Ex: N-dodecyl-alanine

Non-Ionic Surfactants



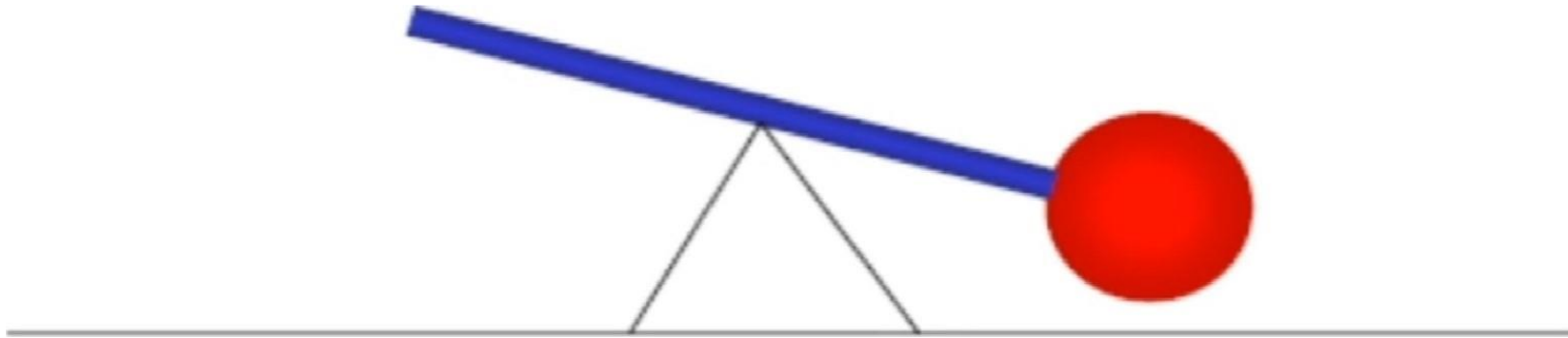
- ✓ Nonionic surfactants comprise the largest group of SAAs that are useful pharmaceutically.
- ✓ The advantages of these agents include low toxicity and high stability and compatibility with both anionic and cationic SAAs
- ✓ Resistance to pH change, and effect of electrolytes.
- ✓ Lower irritancy as compared to other S.A.A.
- ✓ **Ex: Spans (hydrophobic) that form W/O emulsion, Tweens (hydrophilic) that form O/W emulsion**

Depending on the number and nature of the polar and nonpolar groups present, the amphiphile may be hydrophilic, lipophilic or be reasonably well-balanced between these two extremes.

This expresses as Hydrophilic-Lipophilic balance (HLB)

Hydrophilic Lipophilic Balance (HLB)

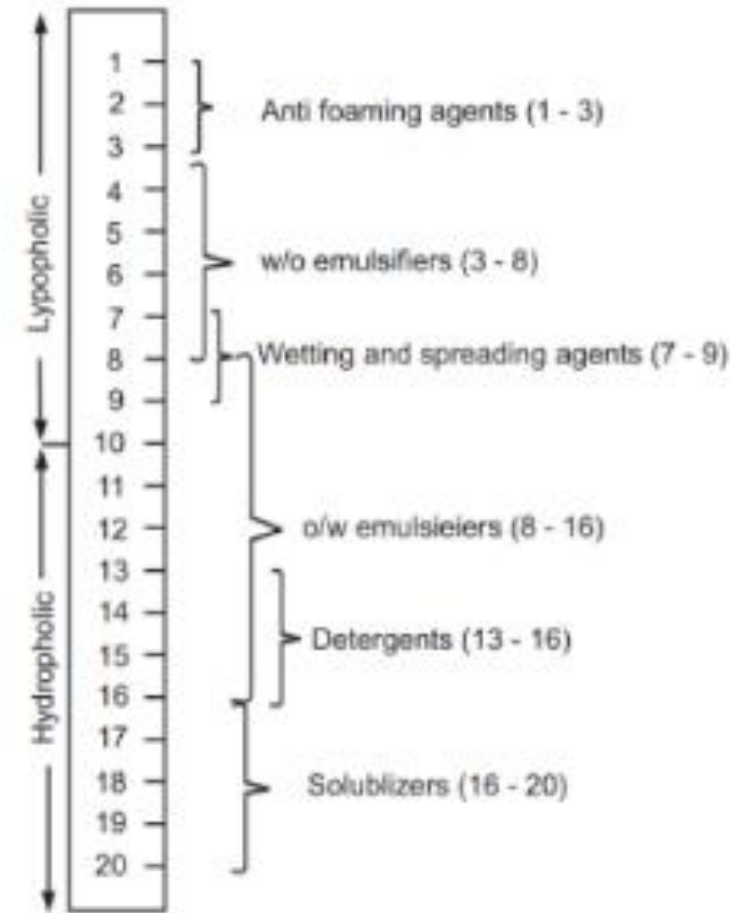
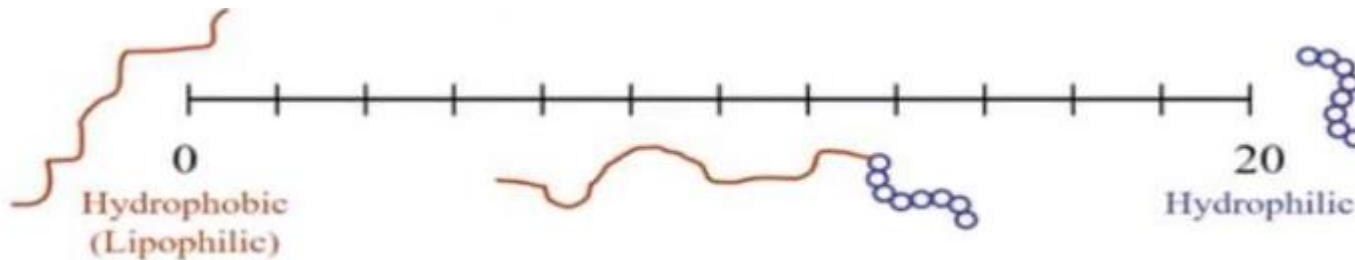
- HLB is a means of expressing **the hydrophilic property of surfactants**



Low HLB values	High HLB values
The lipophilic (hydrophobic) part is dominating	The hydrophilic part is dominating

Griffin' HLB Scale:

- It is an arbitrary scale of values to serve as a measure of the hydrophilic—lipophilic balance (HLB) of surface-active agents.
- HLB ranged from 1-20.
- **Low** HLB (1-9) S.A.A. are **lipophilic**.
- **High** HLB (>9) S.A.A. are **hydrophilic**



- The higher the HLB of a surfactant the more hydrophilic it is.
- Example: *Spans* with low HLB are lipophilic. *Tweens* with high HLB are hydrophilic.

Functional Classification

Griffin' HLB Scale (uses)

- ✓ Indication of the solubility of S.A.A.
- ✓ This system establishes an HLB range of optimum efficiency for each class of surfactant.

HLB Range	Optimum use of SAA
1-3	Anti-foaming
3-8	w/o emulsifying agent
7-10	Wetting agent
8-16	o/w emulsifying agent
12-16	Detergents
16-18	Solubilizing agents

Uses of Surface-Active Agents

- 1- Micelle formation and solubilization
- 2- Emulsifying agents
- 3- Wetting agent
- 4- Detergents
- 5- Foaming and antifoaming agents
- 6- Anti-bacterial agent

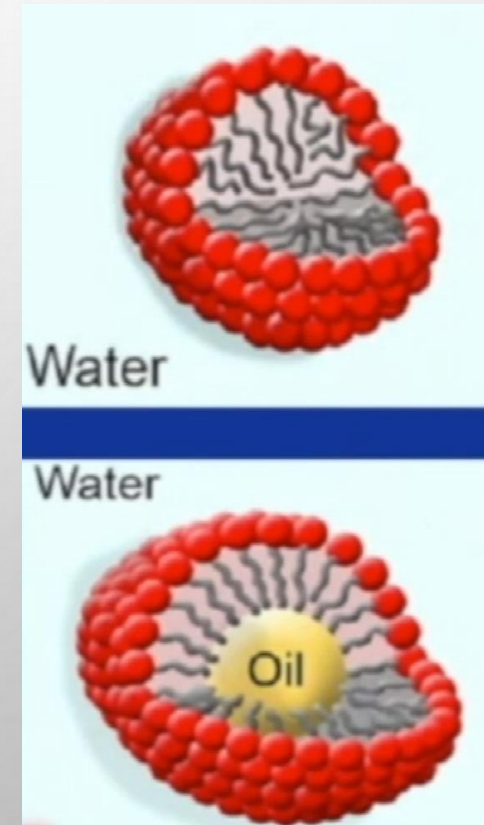
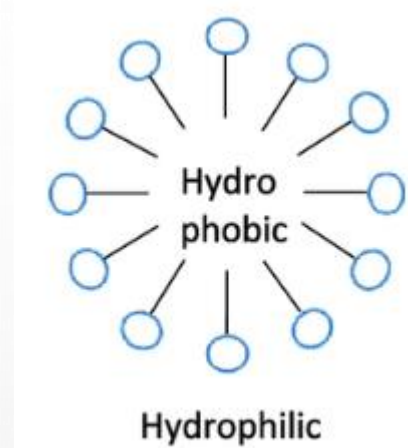
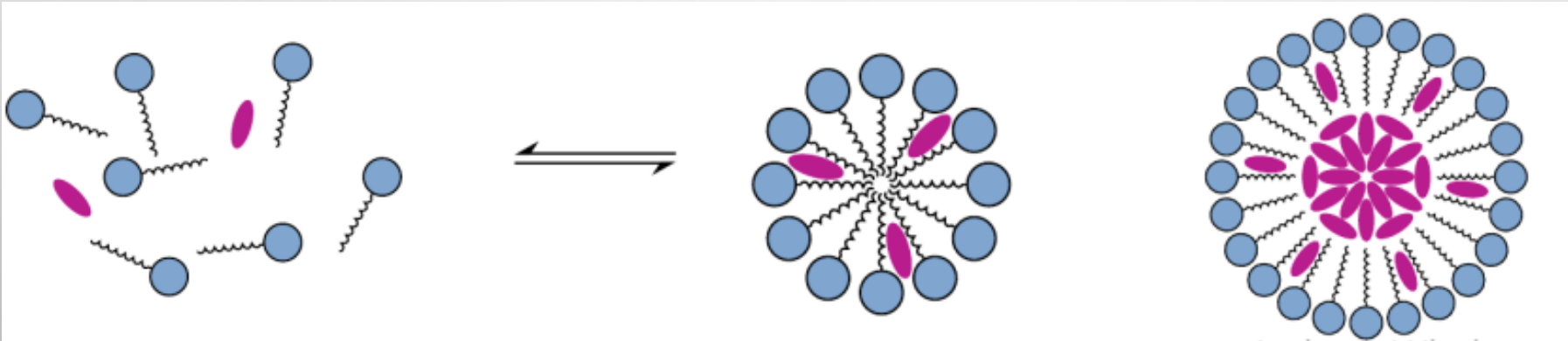
1 - Micelle formation and Solubilization

Solubilization (HLB>16):

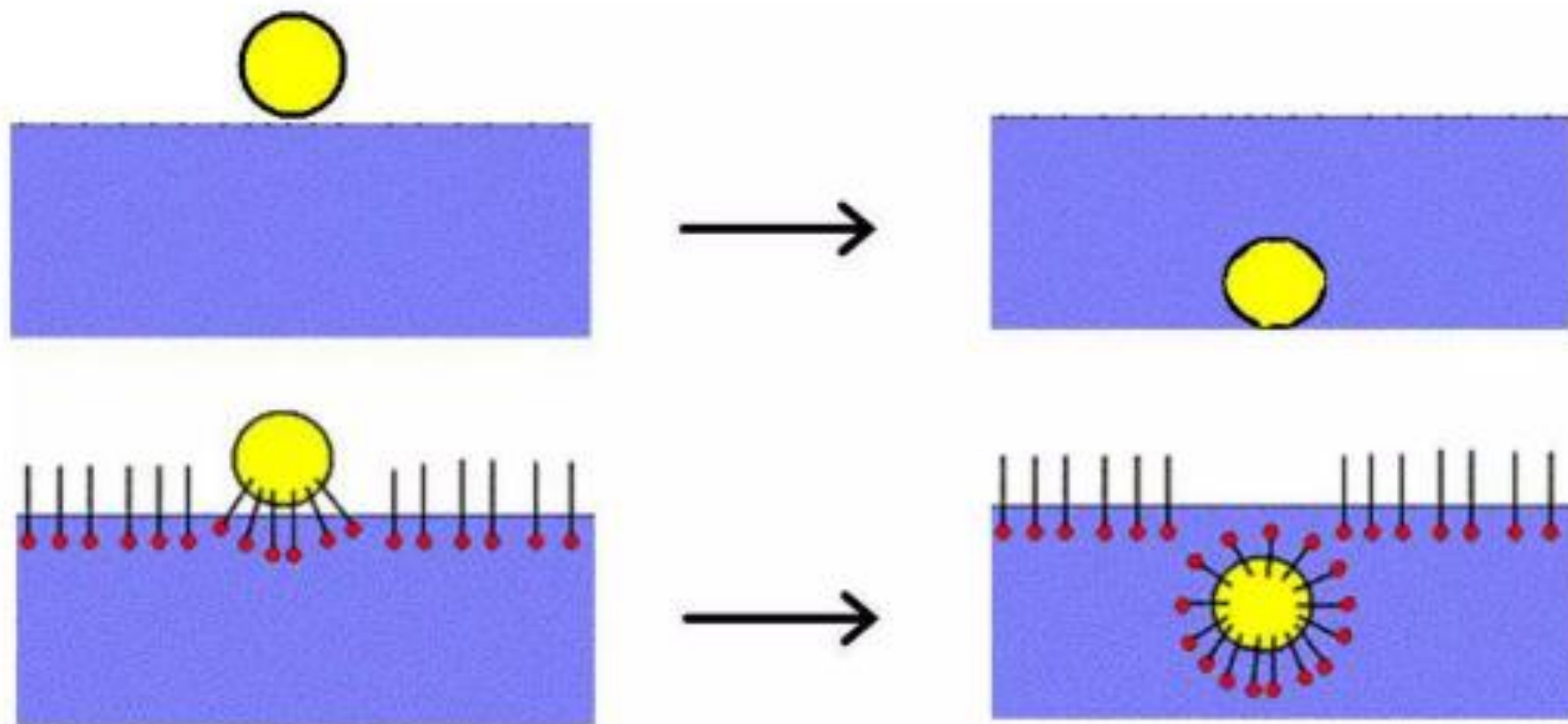
Micellization are essential for Solubilization (Solid in liquid)

Solubilization is the process where water- insoluble substances are brought into solution by incorporation into the hydrophobic core of the micelles forming a molecular dispersion leading to solution.

- Solubilization does not occur until the micelles are formed (i.e. above CMC)
- The amount of substance solubilized increases as the number of micelles increases.



The mechanism for solubilization



Solubilization is the process of incorporation of the solubilize into or onto the micelles.

Pharmaceutical applications of solubilization

A) Phenolic compound:

- Solubilization of cresol with soap to form clear solution for use in disinfection.

Trade name is Lysol

- (50% Cresol+50% soap)

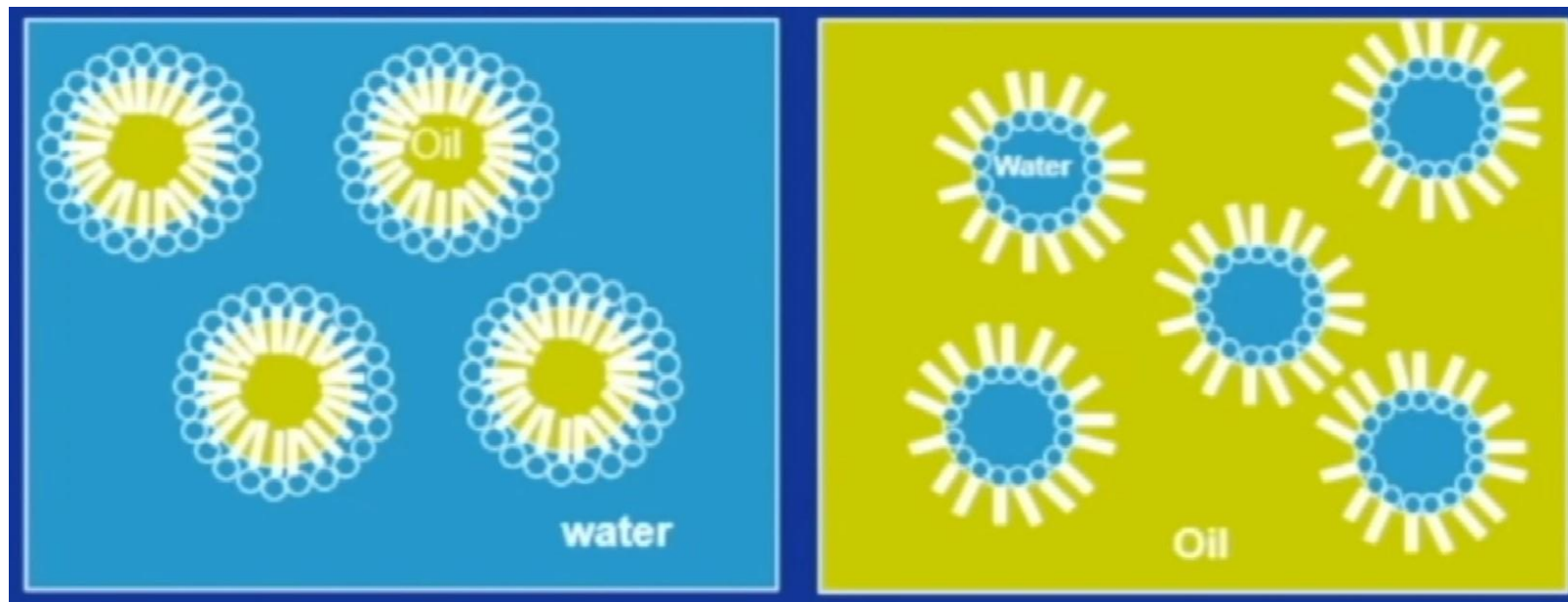
B) Solubilization of some oil soluble drugs

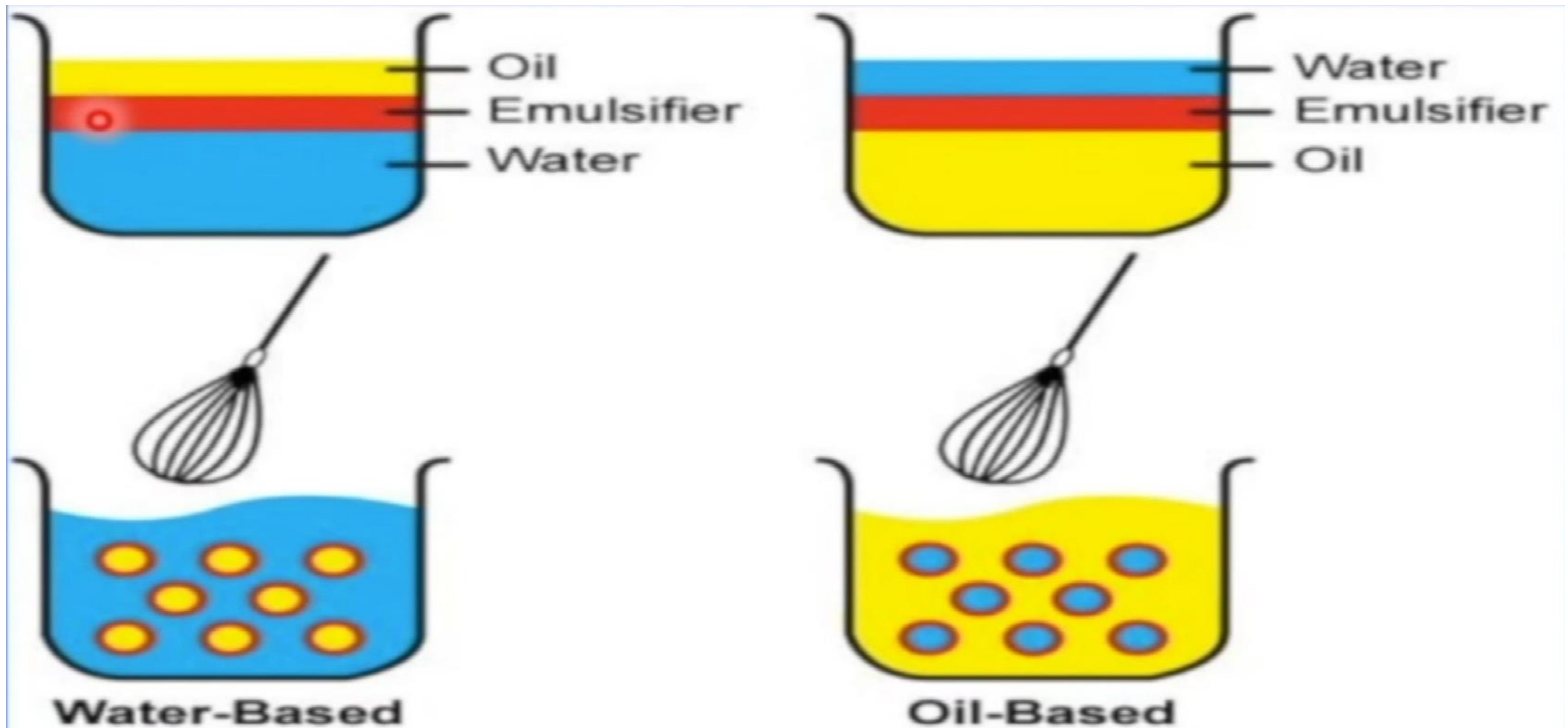
- 1.Oil soluble vitamins (A, D, E, K).
 - 2.Hormonal steroids.
 - 3.Alkaloidal drugs.
 - 4.Glycosides.
- Oil-soluble vitamins such as Vitamin A are unpleasant to take in the form of fish liver oil but are easily palatable when administered in the form of solubilized systems in water

2. Emulsification (Two immiscible liquids)

Based on the HLB value SAA can either emulsify oil drops in a water phase or water droplets in oil phase, leading to the formation of O/W or W/O emulsions, respectively

- Products like creams and lotions are typically emulsions.
- When the product is applied to the skin or hair, the surfactant micelles break open and deliver the oily materials.





o/w emulsion
HLB=8-16

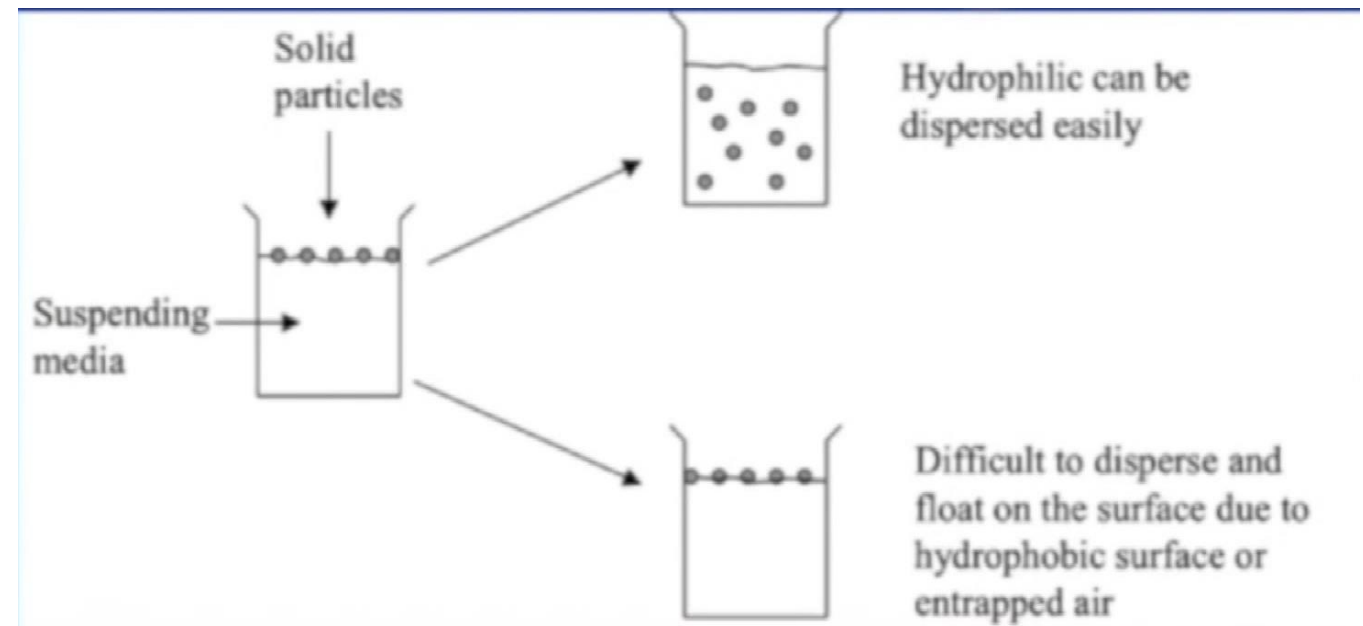
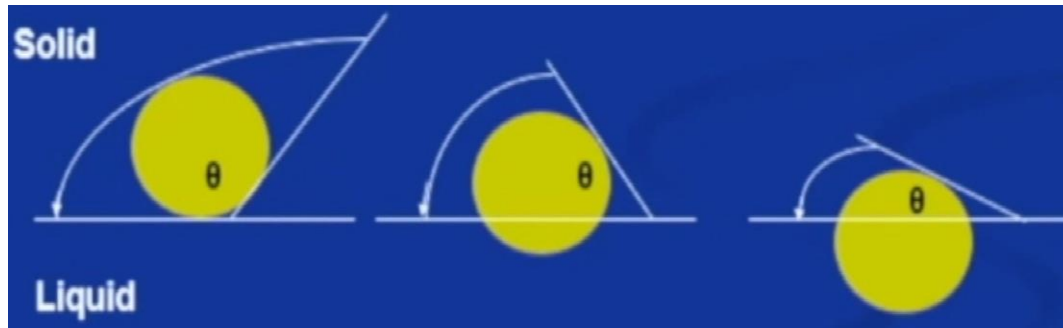
w/o emulsion
HLB=3-8

3. Wetting Effect

HLB=7-10

Wetting refers to the extent of contact between a fluid and a surface, when the two are brought in contact with each other.

- A wetting agent is a surfactant when dissolved in water lowers the advancing contact angle and aids in displacing an air phase at the surface and replacing it with a liquid phase.
- A wetting agent reduces both the surface tension of liquid and the interfacial tension between the solid and liquid. [facilitating good uniform suspension formulation](#)



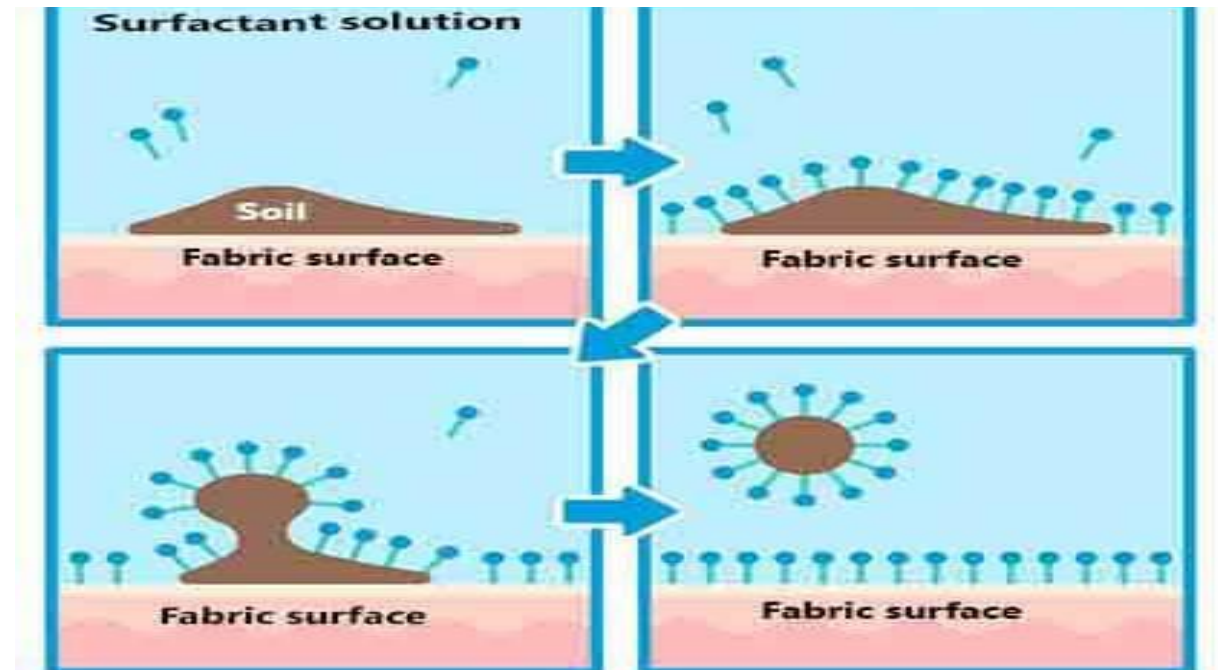
4. Detergents

HLB=12-16

A surface-active agent's detergency is one of its most important properties. The term detergent refers to surface-active agents. Typically, the Detergency term is used when removing or cleaning grease, oil, or dirt from a solid surface. Detergency depends on the formation of micelles. A large part of the process revolves around the action of surfactant molecules.

1. The surfactant needs to have good wetting properties to effectively contact the solid surface.
2. In addition, it can remove dirt from the bulk liquid.

- These are SAA used to remove the dirt from clothes.
- They can be used for the cleansing of body surfaces



5- Foam Formation & Antifoam Effect:

Foaming agents: surfactant that form air globules inside water film, and stabilize them

- ✓ As in toothpaste (high HLB).
- ✓ Foam is opposing a problem in formulations, dispensing and package

Antifoaming agents: S.A.A. that destroy air pockets, they are important in preparations containing S.A.A for wetting effect, they have low HLB (1-3).



THANK YOU!